

Robotics 2

Dynamic model of robots:
Analysis, properties, extensions, uses

Prof. Alessandro De Luca

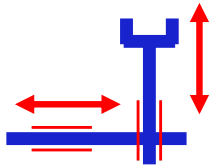
DIPARTIMENTO DI INGEGNERIA INFORMATICA
AUTOMATICA E GESTIONALE ANTONIO RUBERTI



SAPIENZA
UNIVERSITÀ DI ROMA

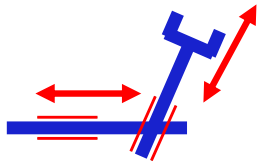
Analysis of inertial couplings

- Cartesian robot



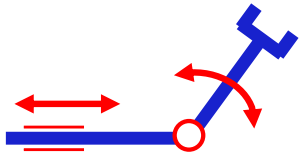
$$M = \begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix}$$

- Cartesian "skew" robot



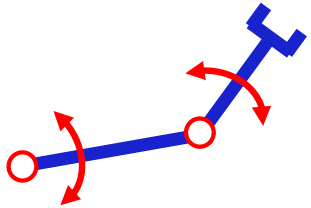
$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{pmatrix}$$

- PR robot



$$M = \begin{pmatrix} m_{11} & m_{12}(q_2) \\ m_{12}(q_2) & m_{22} \end{pmatrix}$$

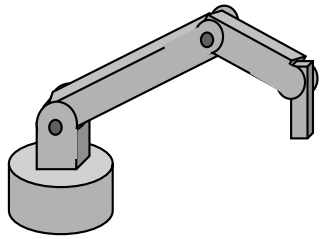
- 2R robot



$$M = \begin{pmatrix} m_{11}(q_2) & m_{12}(q_2) \\ m_{12}(q_2) & m_{22} \end{pmatrix}$$

- 3R articulated robot

(under simplifying assumptions on the locations of the CoMs)



$$M = \begin{pmatrix} m_{11}(q_2, q_3) & 0 & 0 \\ 0 & m_{22}(q_3) & m_{23}(q_3) \\ 0 & m_{23}(q_3) & m_{33} \end{pmatrix}$$

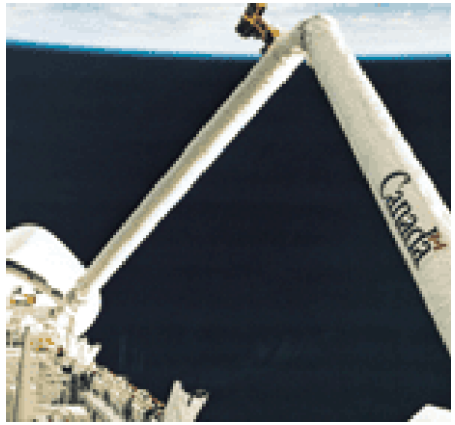
Analysis of gravity term

- absence of gravity
  - constant U_g (motion on horizontal plane)
  - applications in remote space
- static balancing
  - distribution of masses (including motors)
- mechanical compensation
  - articulated system of springs
  - closed kinematic chains

$$g(q) \approx 0$$







Bounds on dynamic terms

- for an open-chain (serial) manipulator, there always exist positive real constants k_0 to k_7 such that, for any value of q and $\dot{q}$

$$k_0 \leq \|M(q)\| \leq k_1 + k_2\|q\| + k_3\|q\|^2 \qquad \text{inertia matrix}$$

$$\|S(q,\dot{q})\| \leq (k_4 + k_5\|q\|)\,\|\dot{q}\| \qquad \text{factorization matrix of Coriolis/centrifugal terms}$$

$$\|g(q)\| \leq k_6 + k_7\|q\| \qquad \text{gravity vector}$$

- if the robot has only **revolute** joints, these simplify to

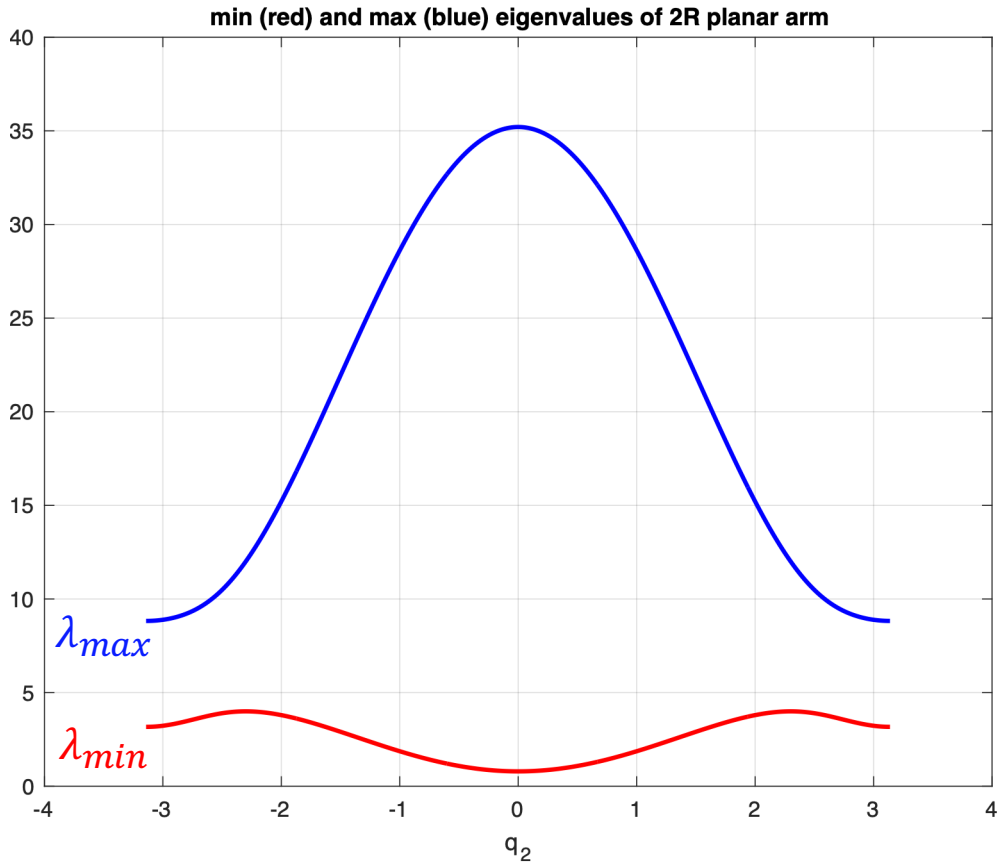
$$k_0 \leq \|M(q)\| \leq k_1 \quad \|S(q,\dot{q})\| \leq k_4\|\dot{q}\| \quad \|g(q)\| \leq k_6$$

(the same holds true with bounds $q_{i,min} \leq q_i \leq q_{i,max}$ on prismatic joints)

NOTE: norms are either for vectors or for matrices (induced norms)

Bounds on inertia matrix

- 2R planar robot with links of uniform density and equal length



inertia matrix

$$M(q) = \begin{pmatrix} 20 + 12\cos q_2 & 4 + 6\cos q_2 \\ 4 + 6\cos q_2 & 4 \end{pmatrix}$$

$$0.79 = k_0 \leq \|M(q)\| \leq k_1 = 35.20$$

p-norm of a vector

$$\|x\|_p = \sqrt[p]{x_1^p + \cdots + x_n^p}$$

2-norm of a vector = Euclidean norm

induced norm of a matrix

$$\|A\| = \sup \frac{\|Ax\|}{\|x\|} \quad x \in \mathbb{R}^n, x \neq 0$$

$$\|A\|_2 = \sigma_{max}(A) = \sqrt{\lambda_{max}(A^T A)}$$

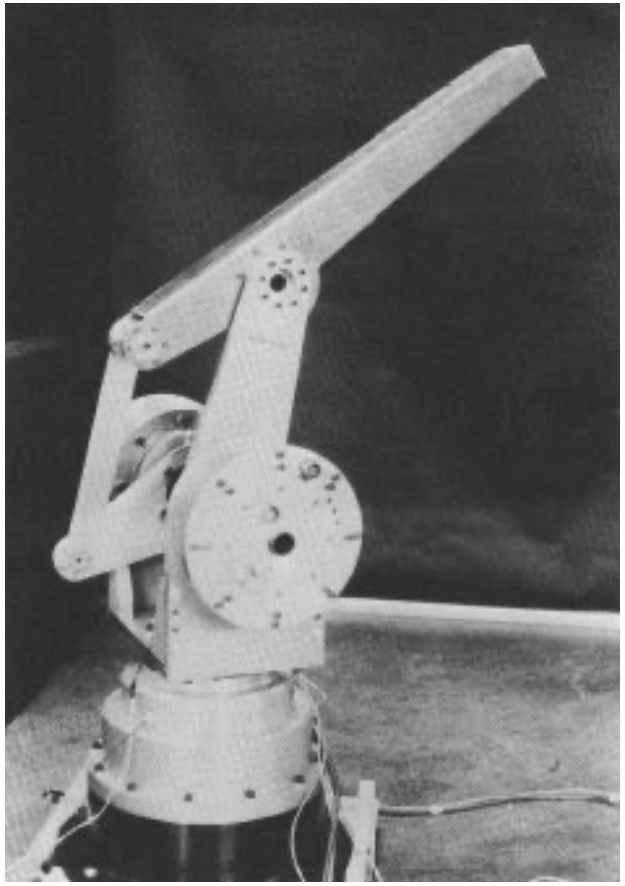
spectral norm

if matrix A is symmetric and positive definite (like $M(q)$) ➡ $\|M\|_2 = \lambda_{max}(M)$

Robots with closed kinematic chains - 1



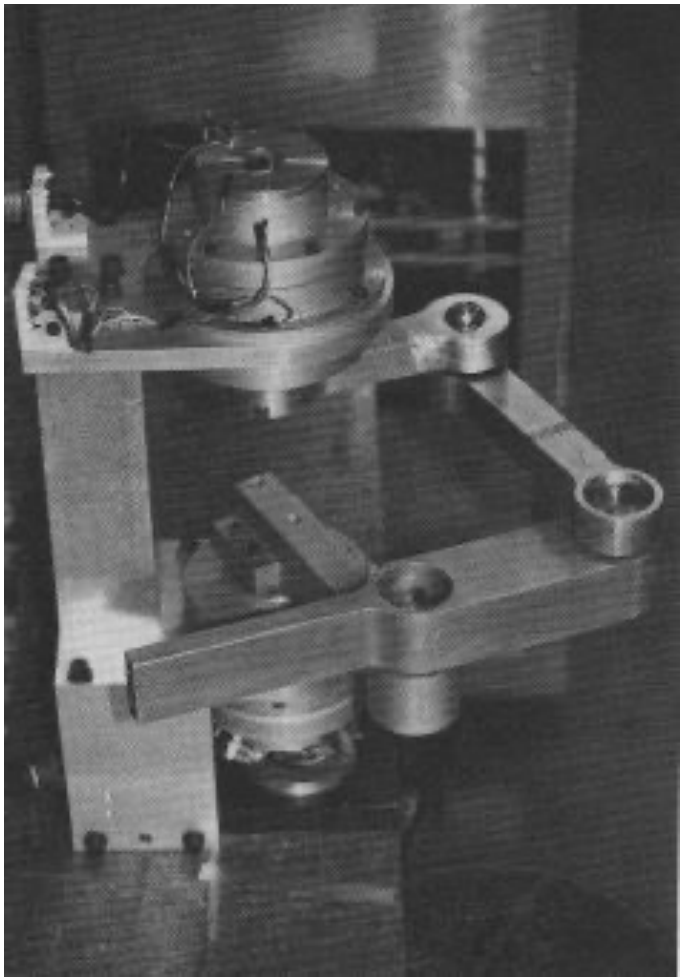
Comau Smart NJ130



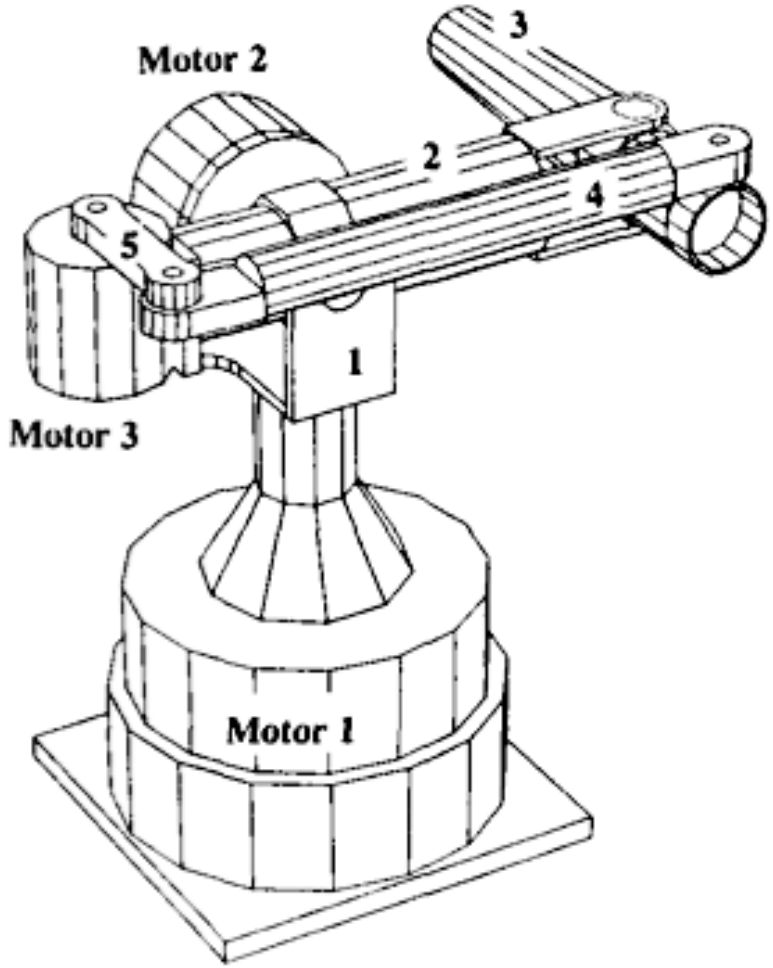


MIT Direct Drive Mark II and Mark III

Robots with closed kinematic chains - 2



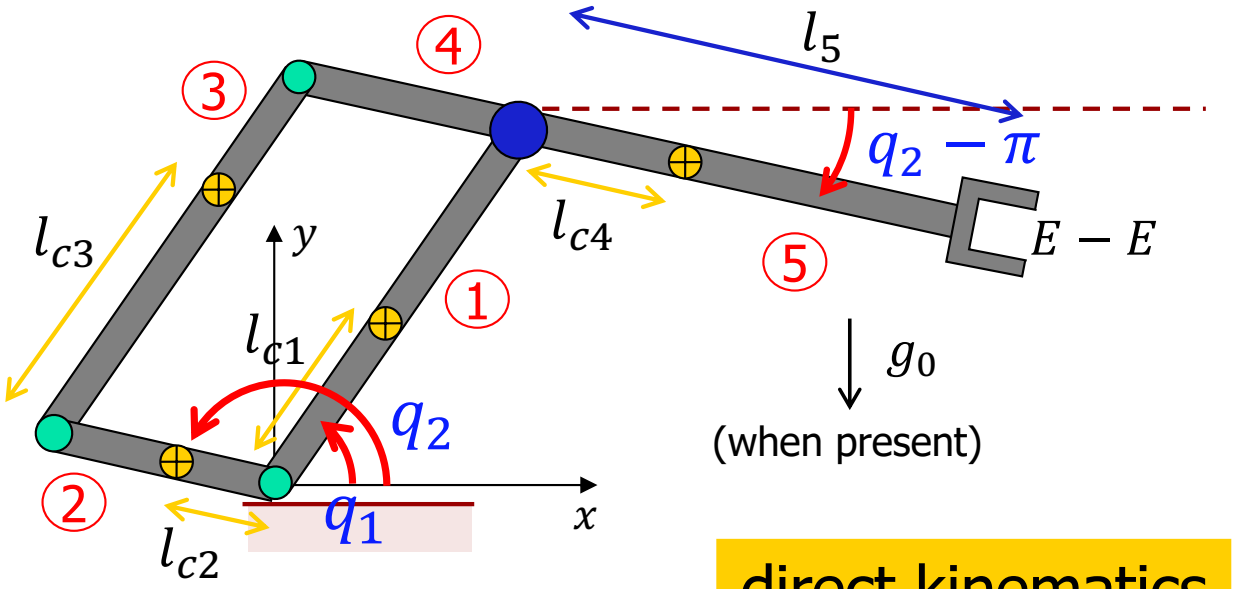
MIT Direct Drive Mark IV
(planar five-bar linkage)



UMinnesota Direct Drive Arm
(spatial five-bar linkage)

Robot with parallelogram structure

(planar) kinematics and dynamics



⊕ center of mass with arbitrary l_{ci}

parallelogram

$$l_1 = l_3$$
$$l_2 = l_4$$

direct kinematics

$$p_{E-E} = \begin{pmatrix} l_1 c_1 \\ l_1 s_1 \end{pmatrix} + \begin{pmatrix} l_5 \cos(q_2 - \pi) \\ l_5 \sin(q_2 - \pi) \end{pmatrix} = \begin{pmatrix} l_1 c_1 \\ l_1 s_1 \end{pmatrix} - \begin{pmatrix} l_5 c_2 \\ l_5 s_2 \end{pmatrix}$$

position of center of masses

$$p_{c1} = \begin{pmatrix} l_{c1} c_1 \\ l_{c1} s_1 \end{pmatrix} \quad p_{c2} = \begin{pmatrix} l_{c2} c_2 \\ l_{c2} s_2 \end{pmatrix} \quad p_{c3} = \begin{pmatrix} l_2 c_2 \\ l_2 s_2 \end{pmatrix} + \begin{pmatrix} l_{c3} c_1 \\ l_{c3} s_1 \end{pmatrix} \quad p_{c4} = \begin{pmatrix} l_1 c_1 \\ l_1 s_1 \end{pmatrix} - \begin{pmatrix} l_{c4} c_2 \\ l_{c4} s_2 \end{pmatrix}$$

a planar (2D) notation is used for position vectors

Kinetic energy

linear/angular velocities

$$v_{c1} = \begin{pmatrix} -l_{c1}s_1 \\ l_{c1}c_1 \end{pmatrix} \dot{q}_1 \qquad v_{c3} = \begin{pmatrix} -l_{c3}s_1 \\ l_{c3}c_1 \end{pmatrix} \dot{q}_1 + \begin{pmatrix} -l_2 s_2 \\ l_2 c_2 \end{pmatrix} \dot{q}_2 \qquad \omega_1 = \omega_3 = \dot{q}_1$$

$$v_{c2} = \begin{pmatrix} -l_{c2}s_2 \\ l_{c2}c_2 \end{pmatrix} \dot{q}_2 \qquad v_{c4} = \begin{pmatrix} -l_1 s_1 \\ l_1 c_1 \end{pmatrix} \dot{q}_1 + \begin{pmatrix} l_{c4}s_2 \\ -l_{c4}c_2 \end{pmatrix} \dot{q}_2 \qquad \omega_2 = \omega_4 = \dot{q}_2$$

a planar notation is used also for velocities: $v_{ci} \in \mathbb{R}^2, \ \omega_i \in \mathbb{R}$

T_i

$$T_1 = \frac{1}{2} m_1 l_{c1}^2 \dot{q}_1^2 + \frac{1}{2} I_{c1,zz} \dot{q}_1^2 \qquad T_2 = \frac{1}{2} m_2 l_{c2}^2 \dot{q}_2^2 + \frac{1}{2} I_{c2,zz} \dot{q}_2^2$$

$$T_3 = \frac{1}{2} m_3 (l_2^2 \dot{q}_2^2 + l_{c3}^2 \dot{q}_1^2 + 2 l_2 l_{c3} c_{2-1} \dot{q}_1 \dot{q}_2) + \frac{1}{2} I_{c3,zz} \dot{q}_1^2$$

$$T_4 = \frac{1}{2} m_4 (l_1^2 \dot{q}_1^2 + l_{c4}^2 \dot{q}_2^2 - 2 l_1 l_{c4} c_{2-1} \dot{q}_1 \dot{q}_2) + \frac{1}{2} I_{c4,zz} \dot{q}_2^2$$

Robot inertia matrix

$$T = \sum_{i=1}^{4} T_i = \frac{1}{2}\dot{q}^T M(q)\dot{q}$$

$$M(q) = \begin{pmatrix} I_{c1,zz} + m_1 l_{c1}^2 + I_{c3,zz} + m_3 l_{c3}^2 + m_4 l_1^2 & \text{symm} \\ (m_3 l_2 l_{c3} - m_4 l_1 l_{c4}) c_{2-1} & I_{c2,zz} + m_2 l_{c2}^2 + I_{c4,zz} + m_4 l_{c4}^2 + m_3 l_2^2 \end{pmatrix}$$

structural condition in mechanical design

$$m_3 l_2 l_{c3} = m_4 l_1 l_{c4} \quad (*)$$

$M(q)$ diagonal and constant ⇒ centrifugal and Coriolis terms ≡ 0

mechanically DECOUPLED and LINEAR dynamic model (up to the gravity term $g(q)$)

$$\begin{pmatrix} M_{11} & 0 \\ 0 & M_{22} \end{pmatrix} \begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

big advantage for the design of motion control laws!

Potential energy and gravity terms

from the y-components of vectors p_{ci}

U_i

$$U_1 = m_1 g_0 l_{c1} s_1 \qquad U_2 = m_2 g_0 l_{c2} s_2$$
$$U_3 = m_3 g_0 (l_2 s_2 + l_{c3} s_1) \qquad U_4 = m_4 g_0 (l_1 s_1 - l_{c4} s_2)$$

$$U = \sum_{i=1}^{4} U_i$$

$$g(q) = \left(\frac{\partial U}{\partial q}\right)^T = \begin{pmatrix} g_0(m_1 l_{c1} + m_3 l_{c3} + m_4 l_1) c_1 \\ g_0(m_2 l_{c2} + m_3 l_2 - m_4 l_{c4}) c_2 \end{pmatrix} = \begin{pmatrix} g_1(q_1) \\ g_2(q_2) \end{pmatrix}$$

gravity components are always "decoupled"

in addition, when $(*)$ holds

$$m_{11}\ddot{q}_1 + g_1(q_1) = u_1$$
$$m_{22}\ddot{q}_2 + g_2(q_2) = u_2$$

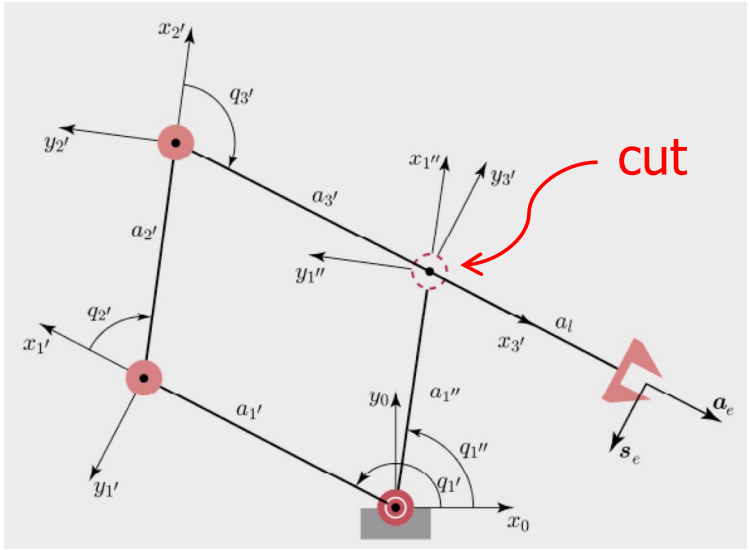
u_i are (nonconservative) torques performing work on q_i

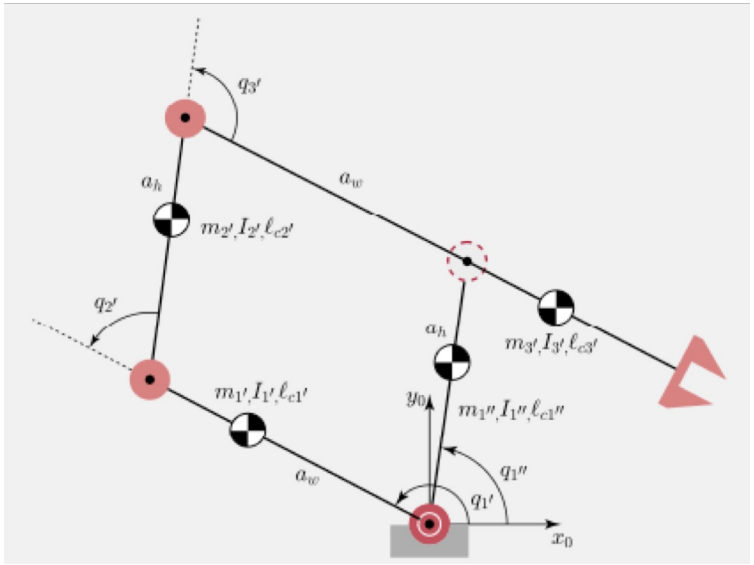
with further structural conditions in the mechanical design: $g(q) \equiv 0$!

Closed kinematic chains

for a more general treatment of simple closed kinematic chains, see textbook

- kinematics: Sect. 2.7.2 introduces a modified DH notation
- select a passive joint in the chain and do a virtual cut to open the chain
- assign DH frames to the two branches, with n joint variables q' and q''
- impose the loop closure as a set of k equality constraints and solve them (numerically or analytically) for a set of $m = n - k$ independent variables
- dynamics: Sect. 5.4.2 follows Euler-Lagrange





Adding dynamic terms ...

1) dissipative phenomena due to friction at the joints/transmissions
   - viscous, Coulomb, stiction ('static friction'), Stribeck, LuGre (dynamic)...
   - local effects at the joints
   - difficult to model in general, except for

$$u_{vi} = -B_{vi}\,\dot{q}_i \qquad u_{ci} = -B_{ci}\,\mathrm{sign}(\dot{q}_i)$$

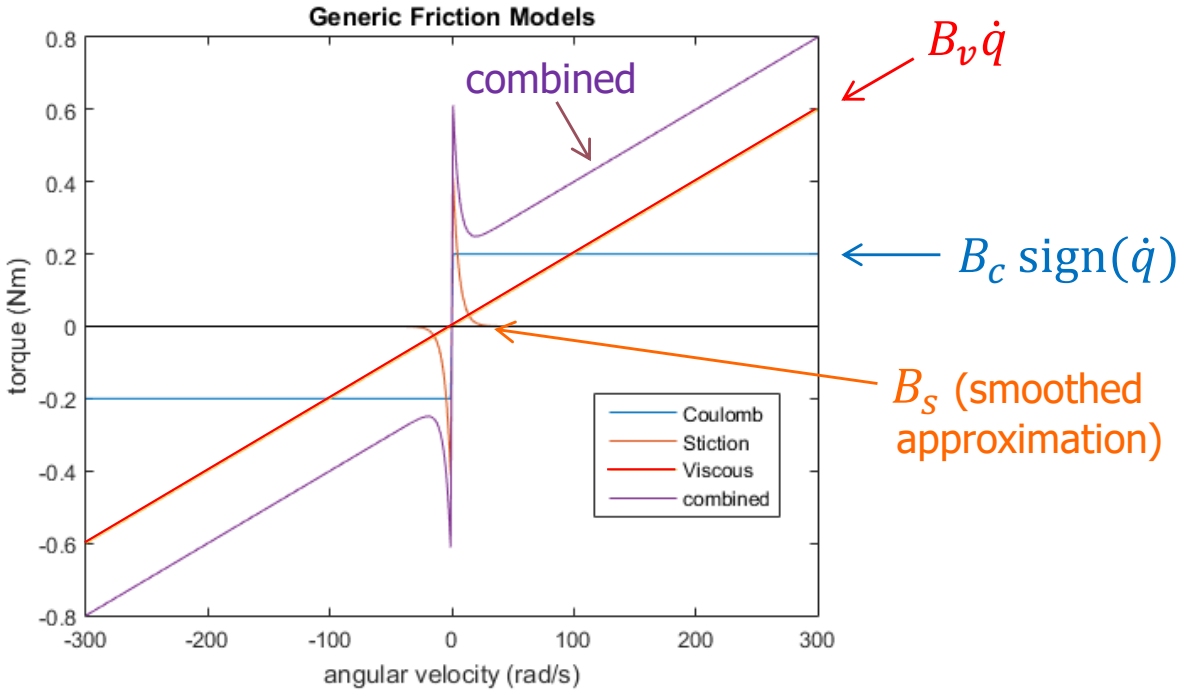
\+ opposes 'intention' of motion only at rest (force to overcome)

$$u_{si} = -B_{si}$$

in general,

$$u_{\mathrm{diss}}^T\,\dot{q} < 0$$

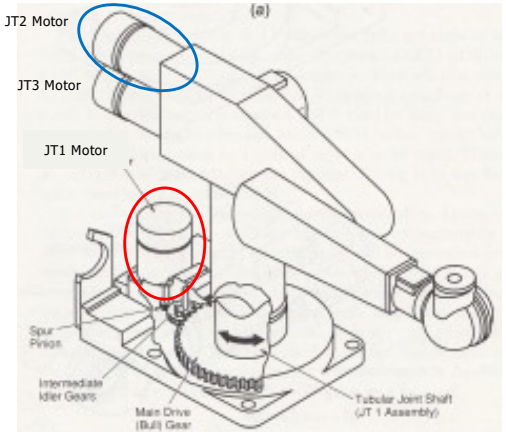
(component-wise too)

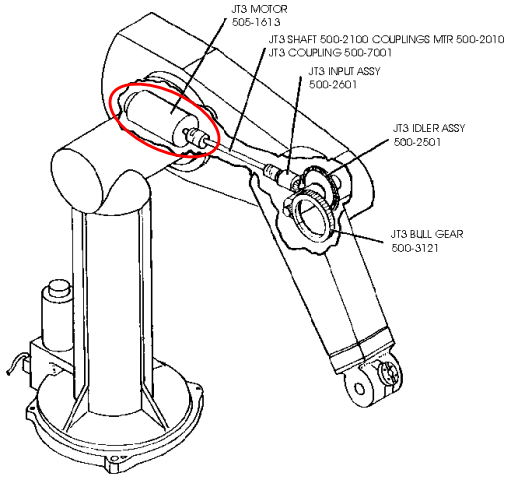


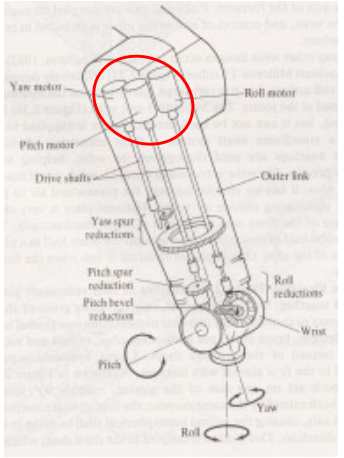
Adding dynamic terms ...

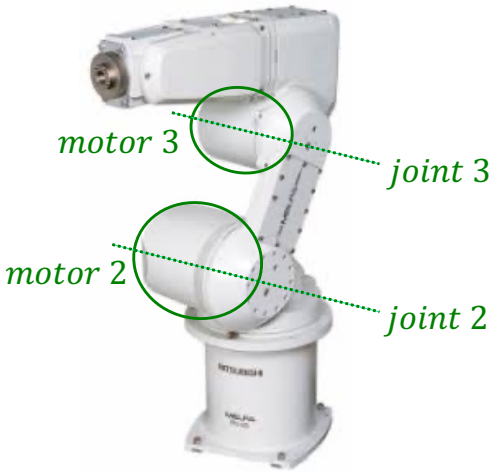
2) inclusion of electrical actuators (as additional rigid bodies)
   - motor i mounted on link $i-1$ (or before), with very few exceptions
   - often with its spinning axis aligned with joint axis i
   - (balanced) mass of motor included in total mass of carrying link
   - (rotor) inertia is to be added to robot kinetic energy
   - transmissions with reduction gears (often, large reduction ratios)
   - in some cases, multiple motors cooperate in moving multiple links: use a transmission coupling matrix Γ (with off-diagonal elements)

Unimation PUMA family



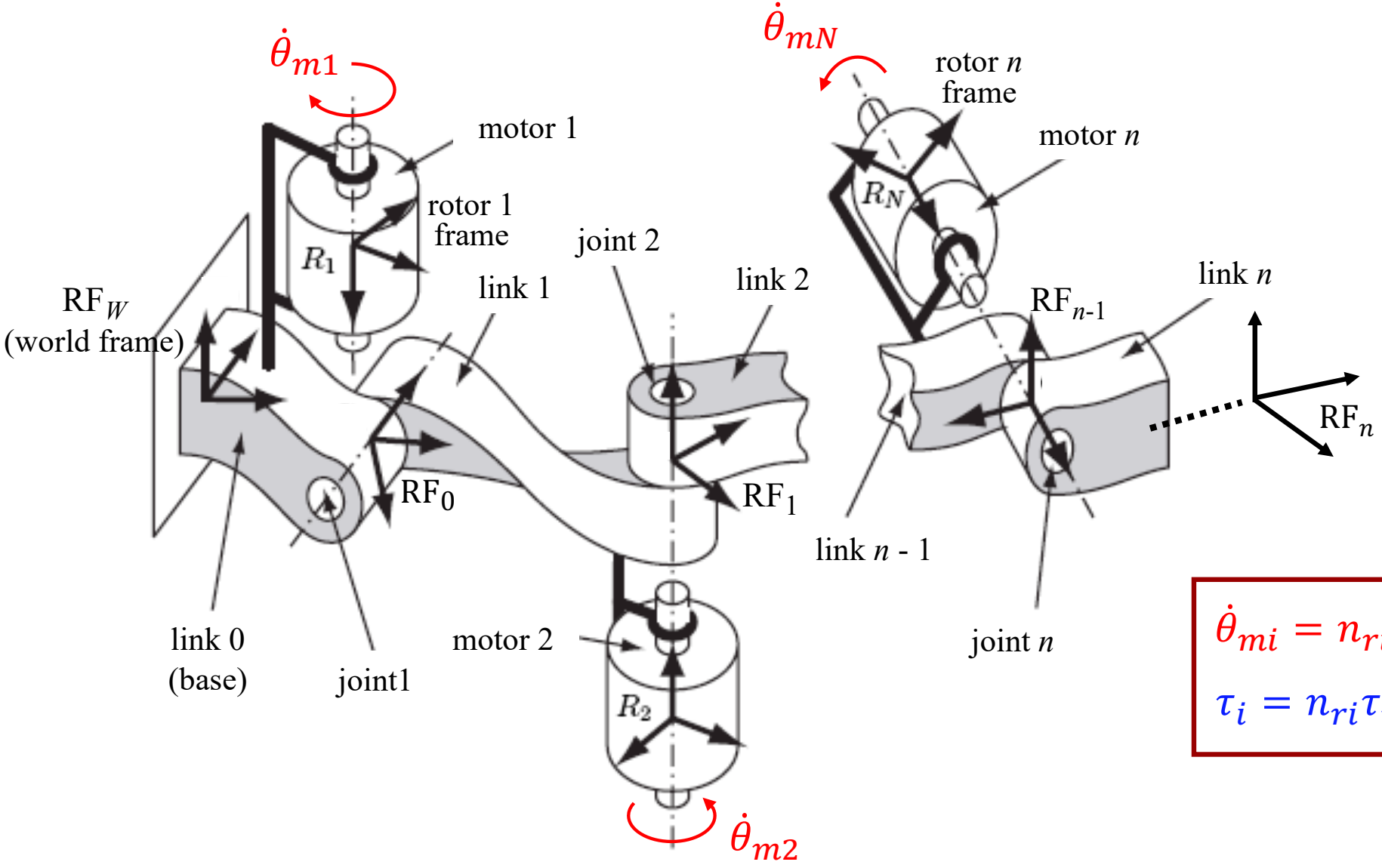






Mitsubishi RV-3S

Placement of motors along the chain



$$\dot{\theta}_{mi} = n_{ri}\dot{\theta}_i$$

$$\tau_i = n_{ri}\tau_{mi}$$

Resulting dynamic model

- simplifying assumption: in the rotational part of the kinetic energy, only the "spinning" rotor velocity is considered

$$T_{mi} = \frac{1}{2} I_{mi} \dot{\theta}_{mi}^2 = \frac{1}{2} I_{mi} n_{ri}^2 \dot{q}_i^2 = \frac{1}{2} M_{mi} \dot{q}_i^2 \qquad T_m = \sum_{i=1}^{n} T_{mi} = \frac{1}{2} \dot{q}^T M_m \dot{q}$$

diagonal, > 0

- including all added terms, the robot dynamics becomes

$$(M(q) + M_m)\ddot{q} + c(q, \dot{q}) + g(q) + B_v \dot{q} + B_c \operatorname{sign}(\dot{q}) = \tau$$

moved to the left ...

constant

does NOT contribute to c

$B_v > 0, B_c > 0$ diagonal

motor torques (**after** reduction gears)

- scaling by the reduction gears, looking from the motor side

$$\left(I_m + \operatorname{diag}\left\{ \frac{m_{ii}(q)}{n_{ri}^2} \right\} \right) \ddot{\theta}_m + \operatorname{diag}\left\{ \frac{1}{n_{ri}} \right\} \left(\bar{M}(q)\ddot{q} + n(q, \dot{q}) \right) = \tau_m$$

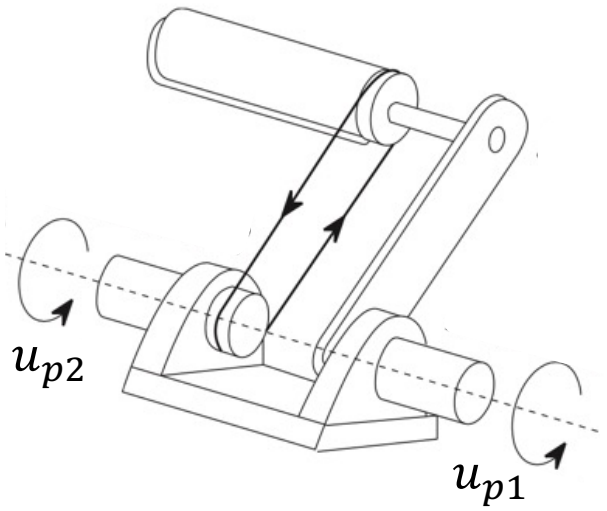
diagonal

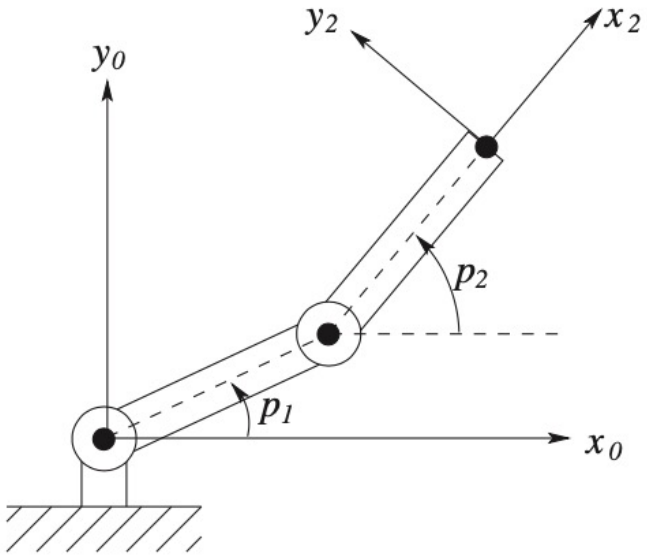
except the diagonal terms m_{jj} $(j = 1, \ldots, n)$

motor torques (**before** reduction gears)

Special actuation and associated coordinates

planar 2R robot with remotely driven forearm





- motor 1 moves link 1 by p_1
- motor 2 at the base moves the absolute angle p_2 of link 2
- derive the dynamic model from scratch using the absolute coordinates $p \in \mathbb{R}^2$

$$M(p)\ddot{p} + c(p,\dot{p}) + g(p) = u_p$$

$$M(p) = \begin{pmatrix} \rho_1 - \rho_3 & \rho_2 c_{2-1} \\ \rho_2 c_{2-1} & \rho_3 \end{pmatrix}$$

$$c(p,\dot{p}) = \begin{pmatrix} -\rho_2 s_{2-1}\, \dot{p}_2^2 \\ \rho_2 s_{2-1}\, \dot{p}_1^2 \end{pmatrix}$$

no more Coriolis torques!

$$g(p) = \begin{pmatrix} \rho_4 c_1 \\ \rho_5 c_2 \end{pmatrix}$$

$c_1 = \cos p_1 \quad c_2 = \cos p_2$

$c_{2-1} = \cos(p_2 - p_1) \quad s_{2-1} = \sin(p_2 - p_1)$

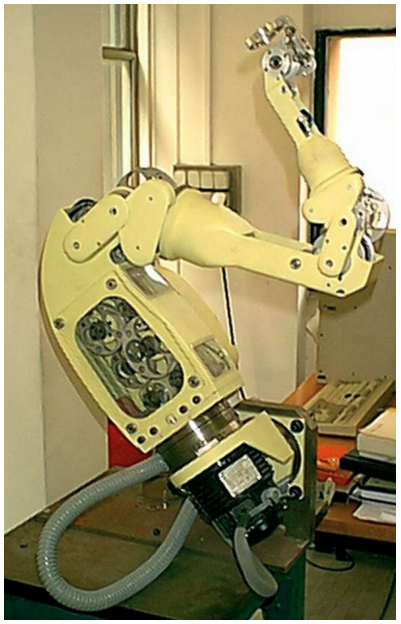
Including joint elasticity



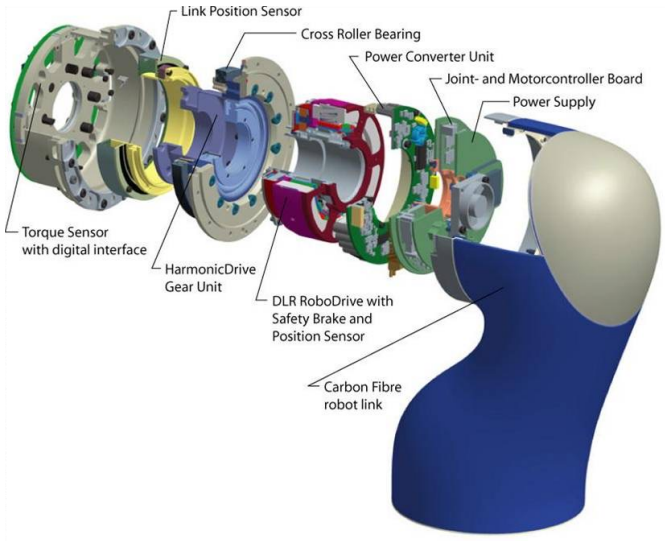
- in industrial robots, use of motion transmissions based on
  - belts
  - harmonic drives
  - long shafts

  introduces flexibility between actuating motors (input) and driven links (output)
- in research robots, compliance in transmissions is introduced on purpose for safety (human collaboration) and/or energy efficiency
  - actuator relocation by means of (compliant) cables and pulleys
  - harmonic drives and lightweight (but rigid) link design
  - redundant (macro-mini or parallel) actuation, with elastic couplings
- in both cases, flexibility is modeled as concentrated at the joints
- in most cases, assuming small joint deformation (elastic domain)

Robots with joint elasticity



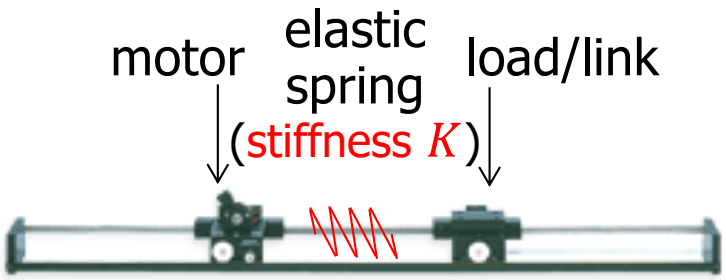
Dexter
with cable transmissions



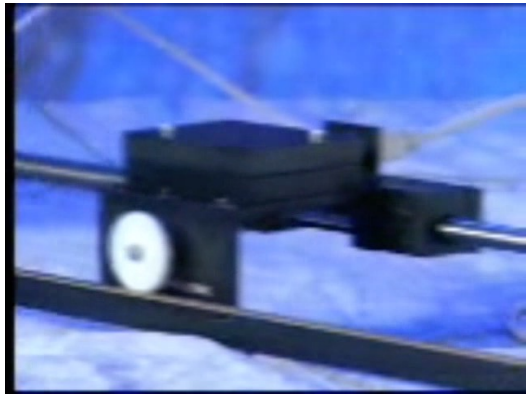
DLR LWR-III
with harmonic drives (HD)



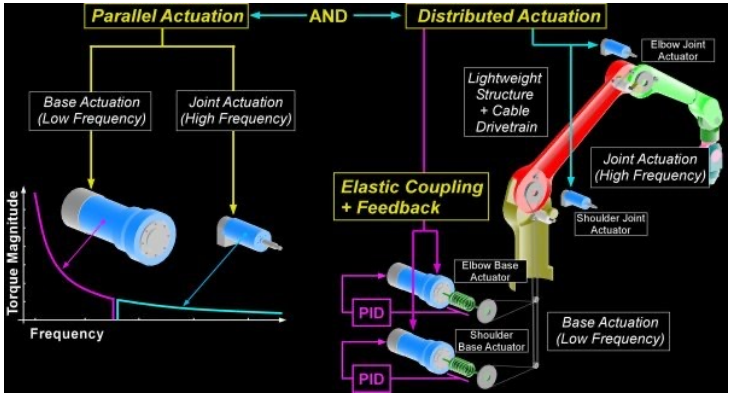
KUKA LBR
iiwa 14 R820
with HD



Quanser Flexible Joint
(1-dof linear, educational)



video (with only motor or
or with full state feedback)



Stanford DECMMA
(with micro-macro actuation)

Dynamic model of robots with elastic joints



- introduce $2n$ generalized coordinates
  - $q = n$ link positions
  - $\theta = n$ motor positions (after reduction, $\theta_i = \theta_{mi}/n_{ri}$)
- add motor kinetic energy T_m to that of the links $T_q = \frac{1}{2}\dot{q}^T M(q)\dot{q}$

$$T_{mi} = \frac{1}{2} I_{mi}\dot{\theta}_{mi}^2 = \frac{1}{2} I_{mi} n_{ri}^2 \dot{\theta}_i^2 = \frac{1}{2} M_{mi}\dot{\theta}_i^2 \qquad T_m = \sum_{i=1}^{n} T_{mi} = \frac{1}{2}\dot{\theta}^T M_m \dot{\theta}$$

diagonal, > 0

- add elastic potential energy U_e to that due to gravity $U_g(q)$
  - $K =$ matrix of joint stiffness

$$U_{ei} = \frac{1}{2} K_i \left(q_i - \left(\frac{\theta_{mi}}{n_{ri}} \right) \right)^2 = \frac{1}{2} K_i (q_i - \theta_i)^2 \qquad U_e = \sum_{i=1}^{n} U_{ei} = \frac{1}{2}(q-\theta)^T K (q-\theta)$$

diagonal, > 0

- apply Euler-Lagrange equations w.r.t. (q, θ)

$2n$ 2nd order differential equations

$$\begin{cases} M(q)\ddot{q} + c(q,\dot{q}) + g(q) + K(q-\theta) = 0 \\ M_m\ddot{\theta} + K(\theta - q) = \tau \end{cases}$$

← no external torques performing work on q if robot is in free space

Use of the dynamic model
inverse dynamics



- given a desired trajectory $q_d(t)$
  - twice differentiable ($\exists\ \ddot{q}_d(t)$)
  - possibly obtained from a task/Cartesian trajectory $y_d(t)$, by (differential) kinematic inversion

  the input torque needed to execute this motion (in free space) is

$$\tau_d = (M(q_d) + M_m)\ddot{q}_d + c(q_d, \dot{q}_d) + g(q_d) + B_v\dot{q}_d + B_c\,\mathrm{sign}(\dot{q}_d)$$

(in contact, with an external wrench) $\ldots - J_{ext}^T(q_d)F_{ext,d}$

- useful also for control (e.g., nominal feedforward)
- however, this way of performing the algebraic computation ($\forall t$) is not efficient when using the Euler-Lagrange modeling approach
  - symbolic terms grow much longer, quite rapidly for larger n
  - in real time, numerical computation is based on Newton-Euler method

State equations
direct dynamics



Euler-Lagrange dynamic model

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) = u$$

n 2nd order differential equations

defining the vector of state variables as $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} q \\ \dot{q} \end{pmatrix} \in \mathbb{R}^{2n}$

state equations

$$\dot{x} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -M^{-1}(x_1)[c(x_1,x_2) + g(x_1)] \end{pmatrix} + \begin{pmatrix} 0 \\ M^{-1}(x_1) \end{pmatrix} u$$

$$= f(x) + G(x)u$$

$2n \times 1$ $\quad$ $2n \times n$

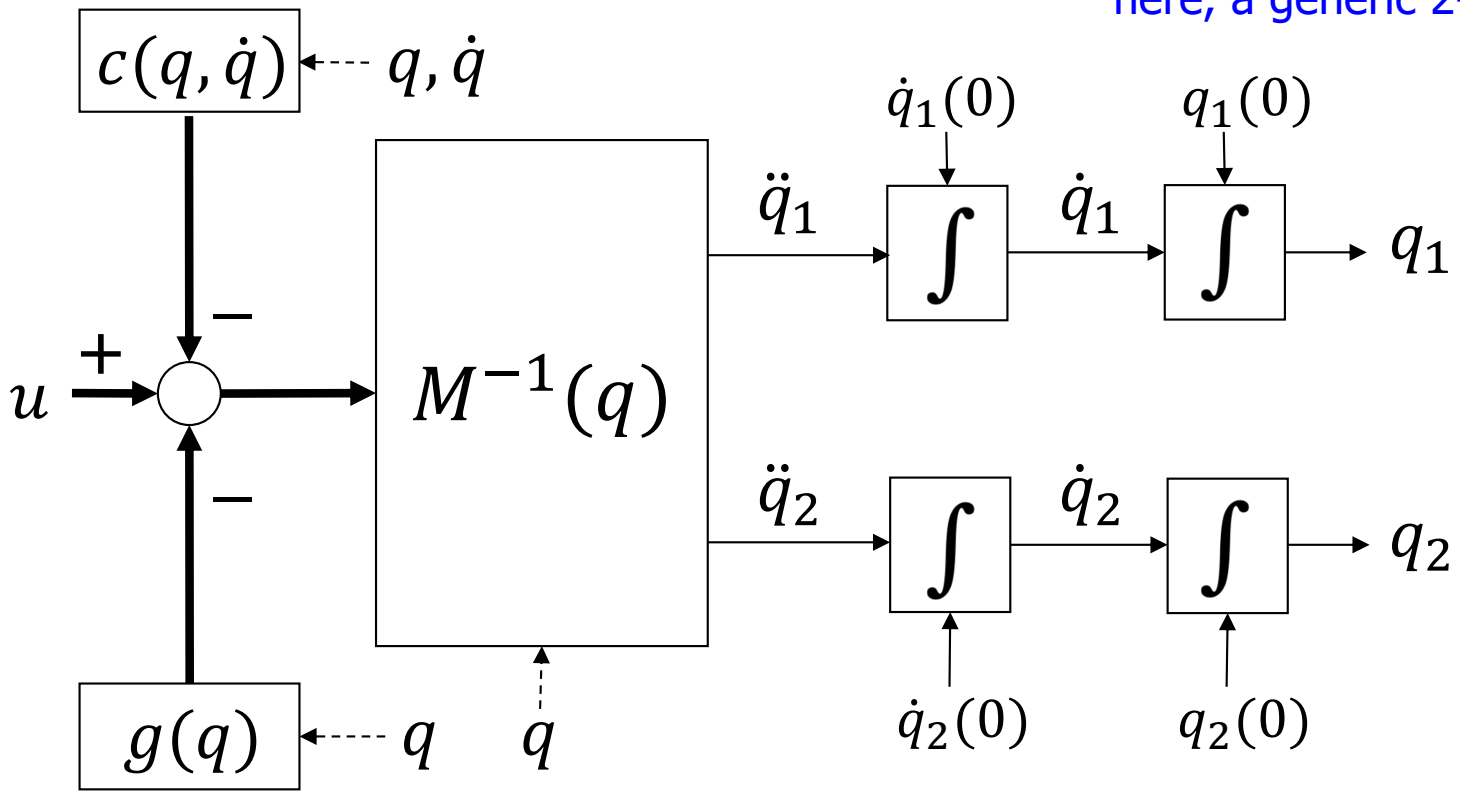
$2n$ 1st order differential equations

another choice... $\tilde{x} = \begin{pmatrix} q \\ M(q)\dot{q} \end{pmatrix}$ ← generalized momentum $\quad$ $\dot{\tilde{x}} =$ ... (do it as exercise)

Dynamic simulation

Simulink-like block diagram

here, a generic 2-dof robot



including "inv(M)"

- initialization (dynamic coefficients and initial state)
- calls to (user-defined) MATLAB functions for the evaluation of model terms
- choice of a numerical integration method (and of its parameters)

e.g., 4th-order Runge-Kutta (ode45)

Approximate linearization

- we can derive a linear dynamic model of the robot, which is valid locally around a given operative condition
  - useful for analysis, design, and gain tuning of linear (or of the linear part of) control laws
  - approximation by Taylor series expansion, up to the first order
  - linearization around a (constant) equilibrium state or along a (nominal, time-varying) equilibrium trajectory
  - usually, we work with (nonlinear) state equations; for mechanical systems, it is more convenient to directly use the 2nd order model
    - same result, but easier derivation

equilibrium state $(q, \dot{q}) = (q_e, 0)$ $[\ \ddot{q} = 0\ ]$ ➡ $g(q_e) = u_e$

equilibrium trajectory $(q, \dot{q}) = (q_d(t), \dot{q}_d(t))$ $[\ \ddot{q} = \ddot{q}_d(t)\ ]$

➡ $M(q_d)\ddot{q}_d + c(q_d, \dot{q}_d) + g(q_d) = u_d$

Linearization at an equilibrium state



- variations around an equilibrium state

$$q = q_e + \Delta q \quad \dot{q} = \dot{q}_e + \Delta\dot{q} = \Delta\dot{q} \quad \ddot{q} = \ddot{q}_e + \Delta\ddot{q} = \Delta\ddot{q} \quad u = u_e + \Delta u$$

- keeping into account the quadratic dependence of c terms on velocity (thus, neglected around the zero velocity)

$$M(q_e)\Delta\ddot{q} + g(q_e) + \underbrace{\left.\frac{\partial g}{\partial q}\right|_{q=q_e}}_{G(q_e)} \Delta q + o\left(\|\Delta q\|, \|\Delta\dot{q}\|\right) = u_e + \Delta u$$

infinitesimal terms of 2nd or higher order

- in state-space format, with $\Delta x = \begin{pmatrix} \Delta q \\ \Delta\dot{q} \end{pmatrix}$

$$\Delta\dot{x} = \begin{pmatrix} 0 & I \\ -M^{-1}(q_e)G(q_e) & 0 \end{pmatrix} \Delta x + \begin{pmatrix} 0 \\ M^{-1}(q_e) \end{pmatrix} \Delta u = A\,\Delta x + B\,\Delta u$$

Linearization along a trajectory

- variations around an equilibrium trajectory

$$q = q_d + \Delta q \qquad \dot{q} = \dot{q}_d + \dot{\Delta q} \qquad \ddot{q} = \ddot{q}_d + \ddot{\Delta q} \qquad u = u_d + \Delta u$$

- developing to 1st order the terms in the dynamic model ...

$$M(q_d + \Delta q)\left(\ddot{q}_d + \ddot{\Delta q}\right) + c(q_d + \Delta q, \dot{q}_d + \dot{\Delta q}) + g(q_d + \Delta q) = u_d + \Delta u$$

$$M(q_d + \Delta q) \cong M(q_d) + \sum_{i=1}^{n} \left.\frac{\partial M_i}{\partial q}\right|_{q=q_d} e_i^T \Delta q$$

e_i^T: i-th row of the identity matrix

$$g(q_d + \Delta q) \cong g(q_d) + G(q_d)\Delta q$$

$$c\left(q_d + \Delta q, \dot{q}_d + \dot{\Delta q}\right) \cong c(q_d, \dot{q}_d) + \overbrace{\left.\frac{\partial c}{\partial q}\right|_{\substack{q=q_d \\ \dot{q}=\dot{q}_d}}}^{C_1(q_d, \dot{q}_d)} \Delta q + \underbrace{\left.\frac{\partial c}{\partial \dot{q}}\right|_{\substack{q=q_d \\ \dot{q}=\dot{q}_d}}}_{C_2(q_d, \dot{q}_d)} \dot{\Delta q}$$

Linearization along a trajectory (cont)



- after simplifications …

$$M(q_d)\,\ddot{\Delta q} + C_2(q_d, \dot{q}_d)\,\dot{\Delta q} + D(q_d, \dot{q}_d, \ddot{q}_d)\,\Delta q = \Delta u$$

with

$$D(q_d, \dot{q}_d, \ddot{q}_d) = G(q_d) + C_1(q_d, \dot{q}_d) + \sum_{i=1}^{N} \left.\frac{\partial M_i}{\partial q}\right|_{q=q_d} \ddot{q}_d e_i^T$$

- in state-space format

$$\dot{\Delta x} = \begin{pmatrix} 0 & I \\ -M^{-1}(q_d)D(q_d, \dot{q}_d, \ddot{q}_d) & -M^{-1}(q_d)C_2(q_d, \dot{q}_d) \end{pmatrix} \Delta x + \begin{pmatrix} 0 \\ M^{-1}(q_d) \end{pmatrix} \Delta u = A(t)\,\Delta x + B(t)\,\Delta u$$

a linear, but time-varying system!

Coordinate transformation

$q \in \mathbb{R}^n$ $\quad M(q)\ddot{q} + c(q,\dot{q}) + g(q) = M(q)\ddot{q} + n(q,\dot{q}) = u_q$ (1)

if we wish/need to use a new set of generalized coordinates p

$p \in \mathbb{R}^n$ $\quad p = f(q) \rightarrow q = f^{-1}(p)$

$\dot{p} = \frac{\partial f}{\partial q}\dot{q} = J(q)\dot{q} \rightarrow \dot{q} = J^{-1}(q)\dot{p}$

$\ddot{p} = J(q)\ddot{q} + \dot{J}(q)\dot{q} \rightarrow \ddot{q} = J^{-1}(q)\left(\ddot{p} - \dot{J}(q)J^{-1}(q)\dot{p}\right)$

by duality (principle of virtual work)

$u_q = J^T(q)u_p$

(1)

$$M(q)J^{-1}(q)\ddot{p} - M(q)J^{-1}(q)\dot{J}(q)J^{-1}(q)\dot{p} + n(q,\dot{q}) = J^T(q)u_p$$

$J^{-T}(q)\cdot$ pre-multiplying the whole equation...

Robot dynamic model
after coordinate transformation



$$J^{-T}(q)M(q)J^{-1}(q)\ddot{p} + J^{-T}(q)\left(n(q,\dot{q}) - M(q)J^{-1}(q)\dot{J}(q)J^{-1}(q)\dot{p}\right) = u_p$$

$q \rightarrow p$

in actual computation, these inner substitutions are not strictly necessary

$(q,\dot{q}) \rightarrow (p,\dot{p})$

nonconservative generalized forces performing work on p

$$M_p(p)\ddot{p} + c_p(p,\dot{p}) + g_p(p) = u_p$$

$$M_p = J^{-T} M\, J^{-1}$$ symmetric, positive definite (out of singularities)

$$g_p = J^{-T} g$$

$$c_p = J^{-T}\left(c - MJ^{-1}\,\dot{J}\,J^{-1}\dot{p}\right) = J^{-T}c - M_p\,\dot{J}\,J^{-1}\dot{p}$$ quadratic dependence on $\dot{p}$

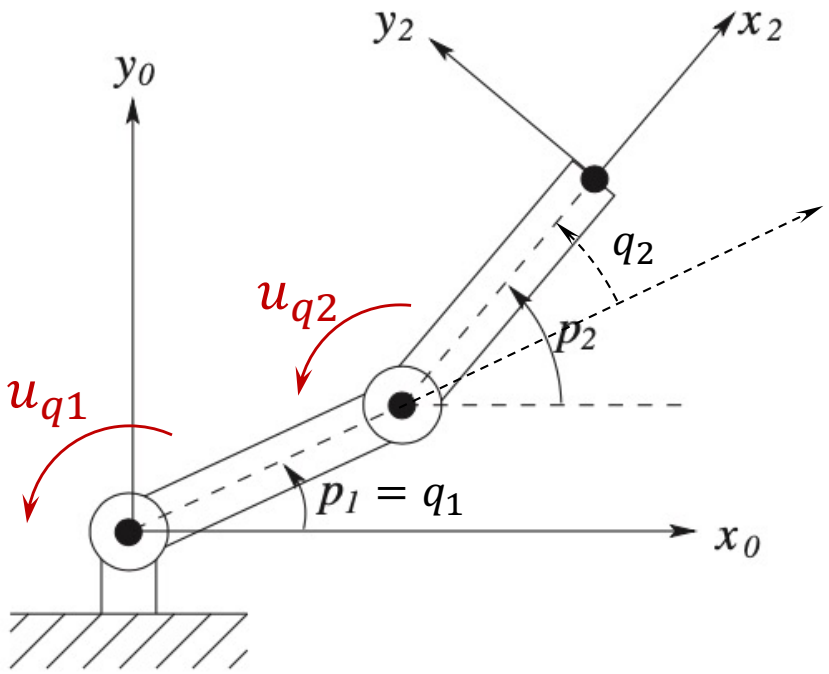
$$c_p(p,\dot{p}) = C_p(p,\dot{p})\,\dot{p}$$

$$\dot{M}_p - 2C_p$$ skew-symmetric

- if $p = y_e$ = E-E pose, this is the robot dynamic model in Cartesian coordinates
  (we have then implicitly assumed that $m = 6 = n$, i.e., no redundancy)

Example of coordinate transformation

planar 2R robot using absolute coordinates



- motor 1 at joint 1, motor 2 at joint 2
- in place of DH angles q, use the absolute angles $p_1 = q_1$ and $p_2 = q_1 + q_2$

$$p = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} q = J\, q$$

a linear transformation

$$J^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \quad J^{-T} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

- from $M(q)\ddot{q} + c(q,\dot{q}) + g(q) = u_q$ obtained with DH relative coordinates

blue terms are the same found in a direct way in slide #17

$$M_p(p) = J^{-T} M\, J^{-1} = \begin{pmatrix} \rho_1 - \rho_3 & \rho_2 c_{2-1} \\ \rho_2 c_{2-1} & \rho_3 \end{pmatrix} \qquad g_p(p) = J^{-T} g = \begin{pmatrix} \rho_4 c_1 \\ \rho_5 c_2 \end{pmatrix}$$

$$c_p(p,\dot{p}) = J^{-T} c = \begin{pmatrix} -\rho_2 s_{2-1}\, \dot{p}_2^2 \\ \rho_2 s_{2-1}\, \dot{p}_1^2 \end{pmatrix} \qquad u_p = J^{-T} u_q = \begin{pmatrix} u_{q1} - u_{q2} \\ u_{q2} \end{pmatrix}$$

Robot dynamic model
in task space, with redundancy



dynamic model in joint space

$$M(q)\ddot{q} + n(q,\dot{q}) = \tau$$

$$q \in \mathbb{R}^n$$
$$y = k(q) \in \mathbb{R}^m$$
$$m < n$$

second-order task kinematics

$$\ddot{y} = J(q)\ddot{q} + \dot{J}(q)\dot{q}$$

J is $m \times n$ with full rank m

1) isolate the joint acceleration from the dynamics ➡ $\ddot{q} = M^{-1}(q)\left(\tau - n(q,\dot{q})\right)$

2) decompose the joint torques $\tau \in \mathbb{R}^n$ in two complementary spaces $\mathcal{R}(J^T) \oplus \mathcal{N}(J^T H)$

$$\tau = J^T(q)F + \left(I - J^T(q)H(q)\right)\tau_N$$

$\in \mathcal{R}(J^T)$ $\quad$ $\in \mathcal{N}(J^T H)$

torques coming from generalized forces F in task space …

H is a generalized inverse of J^T

$$J^T H\, J^T = J^T$$

… and torques $\tau_N \notin \mathcal{R}(J^T)$

➡ $\tau_N = J^T(q)F, \forall F \in \mathbb{R}^m \Rightarrow \left(I - J^T(q)H(q)\right)J^T(q)F = 0$

3) substitute 1) and 2) in the differential task kinematics

➡ $\ddot{y} = J(q)M^{-1}(q)\left(J^T(q)F + (I - J^T(q)H(q))\tau_N - n(q,\dot{q})\right) + \dot{J}(q)\dot{q}$

Robot dynamic model
in task space, with redundancy



4) isolate on the right-hand side the generalized forces F in task space pre-multiplying by $(JM^{-1}J^T)^{-1}$

$$\left(J(q)M^{-1}(q)J^T(q)\right)^{-1}\ddot{y} = F + \left(J(q)M^{-1}(q)J^T(q)\right)^{-1}\left(J(q)M^{-1}(q)\left((I - J^T(q)H(q))\tau_N - n(q,\dot{q})\right) + \dot{J}(q)\dot{q}\right)$$

5) take as generalized inverse the transpose of the inertia-weighted pseudoinverse of the task Jacobian (see block #02 of slides)

$$H(q) = (J(q)M^{-1}(q)J^T(q))^{-1}J(q)M^{-1}(q) = \left(J_M^{\#}(q)\right)^T = (J^T(q))_M^{\#}$$

➡ in this way, the joint torque component τ_N will **never** affect the task acceleration $\ddot{y}$

$$JM^{-1}(I - J^T H)\tau_N = JM^{-1}(I - J^T(JM^{-1}J^T)^{-1}JM^{-1})\tau_N = (JM^{-1} - JM^{-1})\tau_N = 0$$

6) the dynamic equations become

$$\underbrace{\left(J(q)M^{-1}(q)J^T(q)\right)^{-1}}_{M_y(q)}\ddot{y} = F + \underbrace{\left(J(q)M^{-1}(q)J^T(q)\right)^{-1}\left(\dot{J}(q)\dot{q} - J(q)M^{-1}(q)\,n(q,\dot{q})\right)}_{-\,n_y(q)}$$

Robot dynamic model
in task space, with redundancy



7) the resulting m–dimensional task dynamics is

$$M_y(q)\ddot{y} + n_y(q,\dot{q}) = F$$

with

$$M_y(q) = \left(J(q)M^{-1}(q)\,J^T(q)\right)^{-1} \quad \text{task inertia matrix}$$

$$n_y(q,\dot{q}) = M_y(q)\left(J(q)M^{-1}(q)\,n(q,\dot{q}) - \dot{J}(q)\dot{q}\right)$$

- this is called a dynamically consistent formulation since interaction forces F_{int} in the task space can be directly added on the right-hand side of the equations

$$M_y(q)\ddot{y} + n_y(q,\dot{q}) = F + F_{\text{int}}$$

- moreover, for $m = n$, the dynamic terms are identical to those in slide #29 since

$$M_y = (JM^{-1}\,J^T)^{-1} = J^{-T}M\,J^{-1} \quad \Rightarrow \quad M_y\,J\,M^{-1} = J^{-T}$$

8) however, an additional $(n-m)$–dimensional second-order dynamics is needed to describe the complete robot dynamics

Direct and inverse dynamics
options in the presence of task redundancy



- direct dynamics

  a given robot state $(q, \dot{q})$ corresponds to a unique task state $(y, \dot{y})$

  - use $M_y(q)\ddot{y} + n_y(q, \dot{q}) = F$ to compute $\ddot{y}$ from an input task force F
  - if the original input is specified as a joint torque τ, use $F = \left(J_M^{\#}\right)^T \tau$
  - in any case, for simulation purposes, the complete joint dynamics $M(q)\ddot{q} + n(q, \dot{q}) = \tau$ should be used to compute $\ddot{q}$
  - reconstruct then $\ddot{y}$ from $\ddot{q}$ and task kinematics

- inverse dynamics

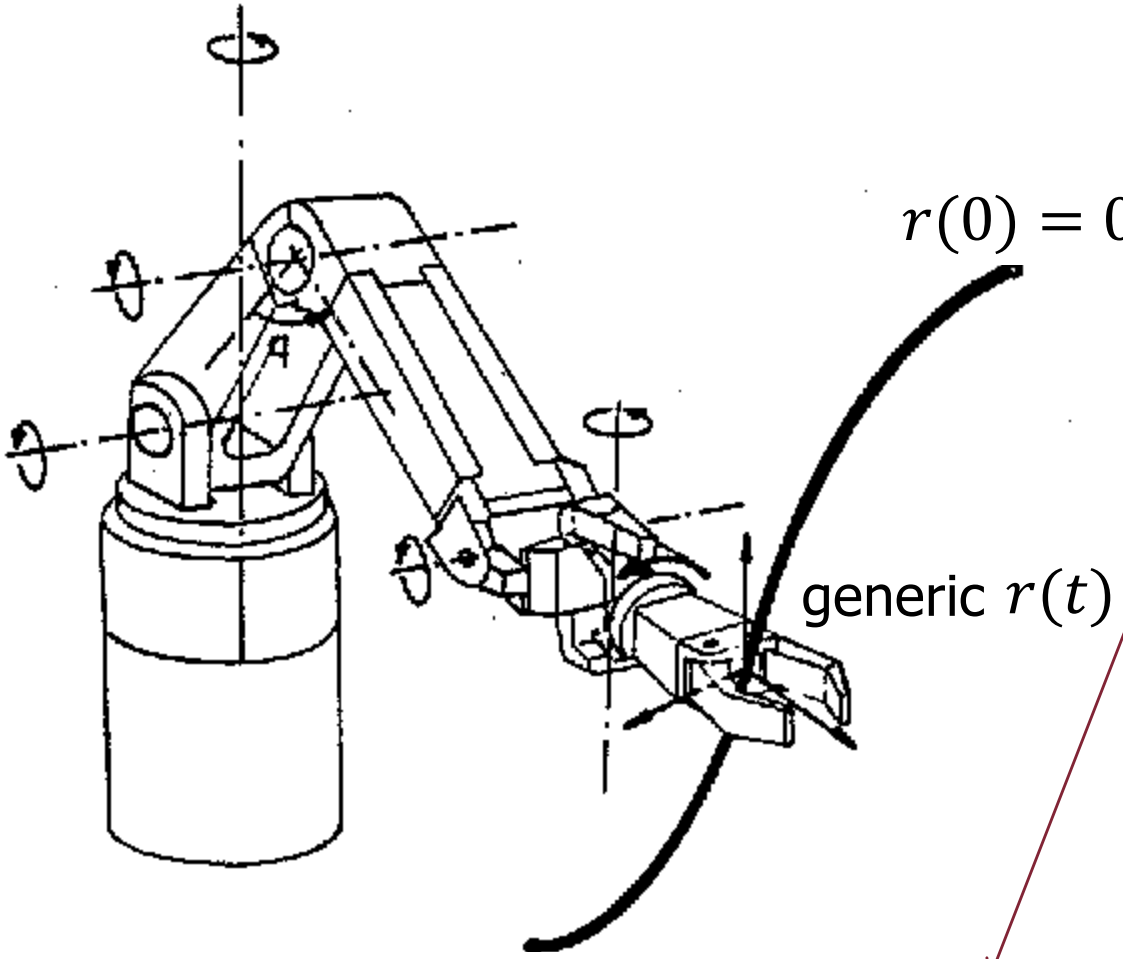
  at a robot state $(q, \dot{q})$, compute the torque τ for a desired $\ddot{y}_d$

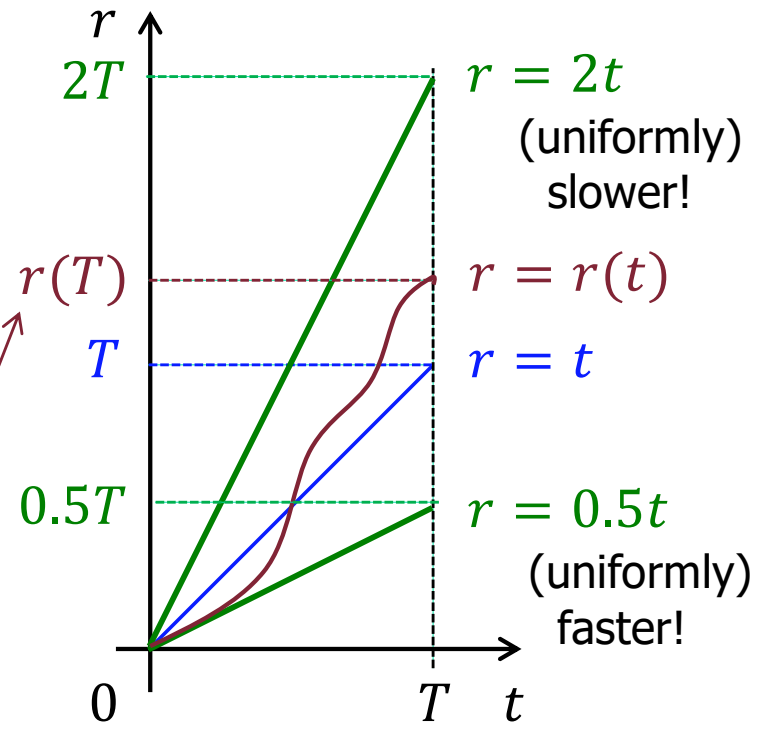
  - to obtain the required task force, use $M_y(q)\ddot{y}_d + n_y(q, \dot{q}) = F_d$
  - all joint torques are given by the dynamically consistent decomposition $\tau = J^T F_d + \left(I - J^T (J^T)_M^{\#}\right)\tau_N = J^T F_d + \left(I - J_M^{\#} J\right)^T \tau_N$, with any τ_N
  - e.g., choosing $\tau_N = 0$ yields the joint torque that minimizes $\frac{1}{2}\|\tau\|_{M^{-1}}^2$

Dynamic scaling of trajectories

uniform time scaling of motion

- given a smooth original trajectory $q_d(t)$ of motion for $t \in [0, T]$
  - assume to rescale time as $t \to r(t)$ (a strictly **increasing** function of t)





Dynamic scaling of trajectories

uniform time scaling of motion



- in the new time scale, the scaled trajectory $q_s(r)$ satisfies

$$q_d(t) = q_s(r(t)) \;\Rightarrow\; \dot{q}_d(t) = \frac{dq_d}{dt} = \frac{dq_s}{dr}\frac{dr}{dt} = q_s'(r)\,\dot{r}(t)$$

same path executed
(at different instants of time)

$$\ddot{q}_d(t) = \frac{d\dot{q}_d}{dt} = \left(\frac{dq_s'}{dr}\frac{dr}{dt}\right)\dot{r} + q_s'\frac{d\dot{r}}{dt} = q_s''(r)\dot{r}^2(t) + q_s'(r)\ddot{r}(t)$$

- uniform scaling of the trajectory occurs when $r(t) = kt$

$$\dot{q}_d(t) = k\,q_s'(kt) \qquad \ddot{q}_d(t) = k^2 q_s''(kt)$$

Q: what is the new input torque needed to execute the scaled trajectory?
(dissipative terms are here neglected in the dynamic model)

Dynamic scaling of trajectories

inverse dynamics under uniform time scaling



- the new torque could be recomputed through the inverse dynamics, for every $r = kt \in [0, T_s] = [0, kT]$ along the scaled trajectory, as

$$\tau_s(kt) = M(q_s)q_s'' + c(q_s, q_s') + g(q_s)$$

- however, being the dynamic model linear in the acceleration and quadratic in the velocity, it is

$$\tau_d(t) = M(q_d)\ddot{q}_d + c(q_d, \dot{q}_d) + g(q_d) = M(q_s)k^2 q_s'' + c(q_s, kq_s') + g(q_s)$$
$$= k^2\big(M(q_s)q_s'' + c(q_s, q_s')\big) + g(q_s) = k^2\big(\tau_s(kt) - g(q_s)\big) + g(q_s)$$

- thus, saving separately total torque $\tau_d(t)$ and gravity torque $g_d(t)$ in the computation of the inverse dynamics along the original trajectory, the new input torque is obtained directly as

$$\tau_s(kt) = \frac{1}{k^2}\big(\tau_d(t) - g(q_d(t))\big) + g(q_d(t))$$

$k > 1$: slow down ⇒ reduce torque

$k < 1$: speed up ⇒ increase torque

the gravity torque (which is only position-dependent) does NOT scale!

Recovering torque feasibility

using uniform time scaling



- let the robot torques delivered by the motors be bounded as

$$|\tau_i| \leq \tau_{i,\max} \qquad i = 1, \dots, n$$

- assume that the maximum torques are at least sufficient to sustain the gravity load at the joints

$$\tau_{i,\max} > |g_i(q)| \qquad \forall q,\ i = 1, \dots, n$$

- if the original trajectory $q_d(t)$ is dynamically unfeasible

$$\exists\, j \in \{1, \dots, n\}, \exists\, \bar{t} \in [0, T] : \left|\tau_{d,j}(q_d(\bar{t}))\right| > \tau_{j,\max}$$

at least one joint is violating in at least one instant

then there exists a minimum scaling factor $k^* > 1$ that provides a feasible scaled trajectory $q_s(k^* t)$ for all $t \in [0, T]$

- compute for $i = 1, \dots, n$ the normalized torque violations (w/o gravity)

$$k_i = \max\left\{1, \max_{t\in[0,T]}\left\{\frac{\tau_i(t) - g_i(q(t))}{\tau_{i,max} - g_i(q(t))}, \frac{g_i(q(t)) - \tau_i(t)}{\tau_{i,max} + g_i(q(t))}\right\}\right\} \geq 1$$

$k_i = 1$ means that joint i is not violating its bounds

- then

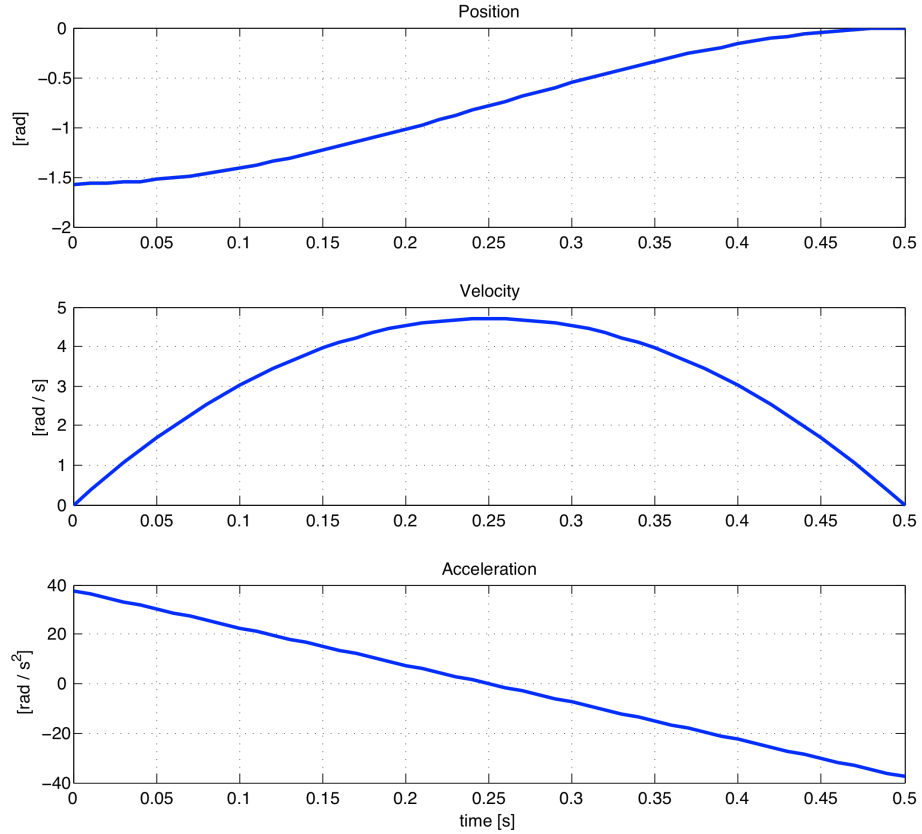
$$k^* = \max_{i=1,\dots,n} \sqrt{k_i} = \sqrt{k_{i^*}} > 1$$

$k^* > 1$ since at least joint i^* is violating one of its bounds

$$T_s = k^* T > T$$

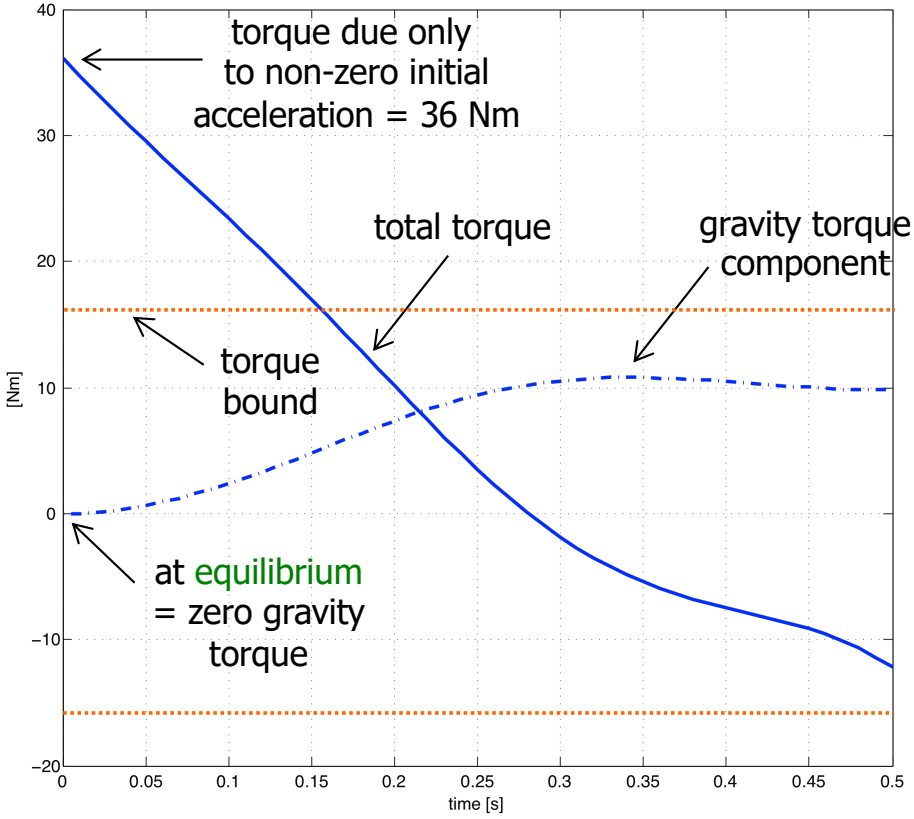
Dynamic scaling of trajectories
numerical example

- rest-to-rest motion with cubic polynomials for planar 2R robot under gravity (from downward equilibrium to horizontal link 1 & upward vertical link 2)
- original trajectory with $T = 0.5$ s violates torque bound at joint 1 ($\tau_{1,\max} = 16$ Nm)



for both joints

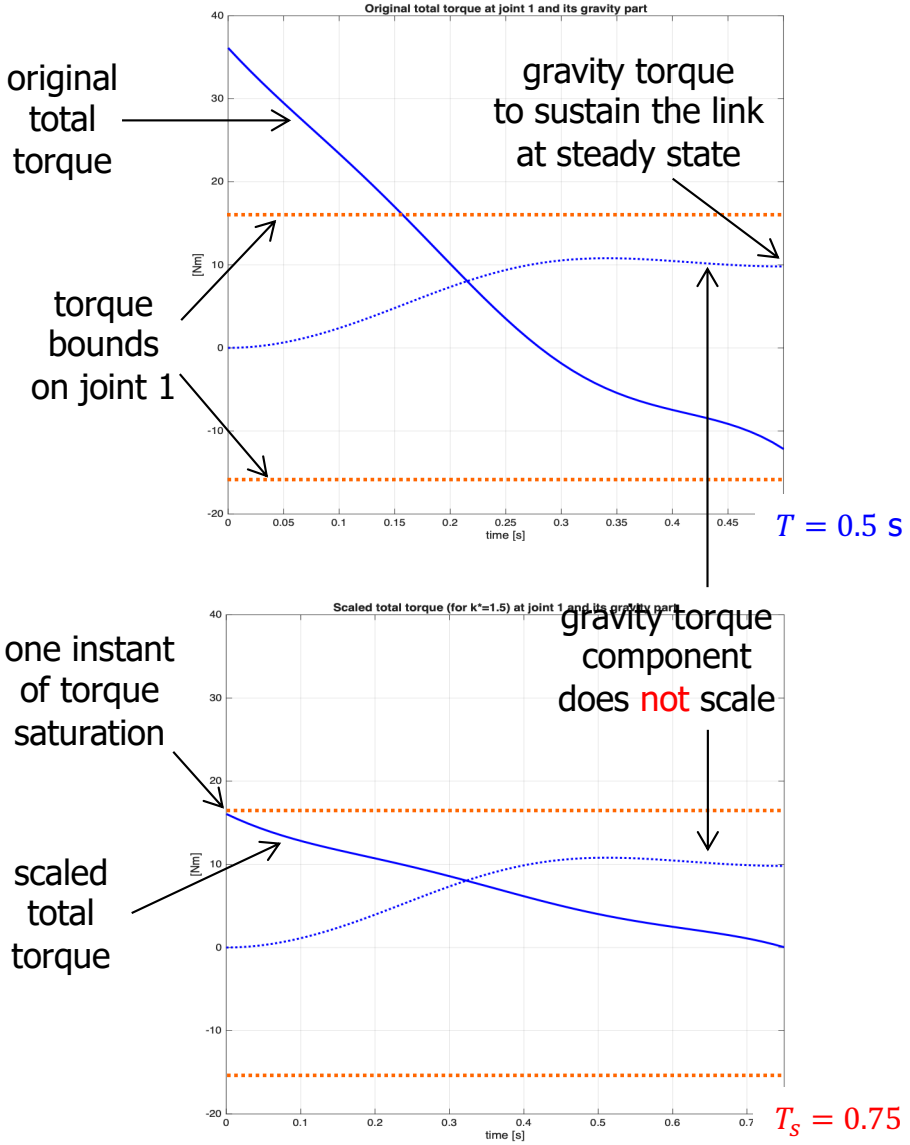
only joint 1 torques are shown



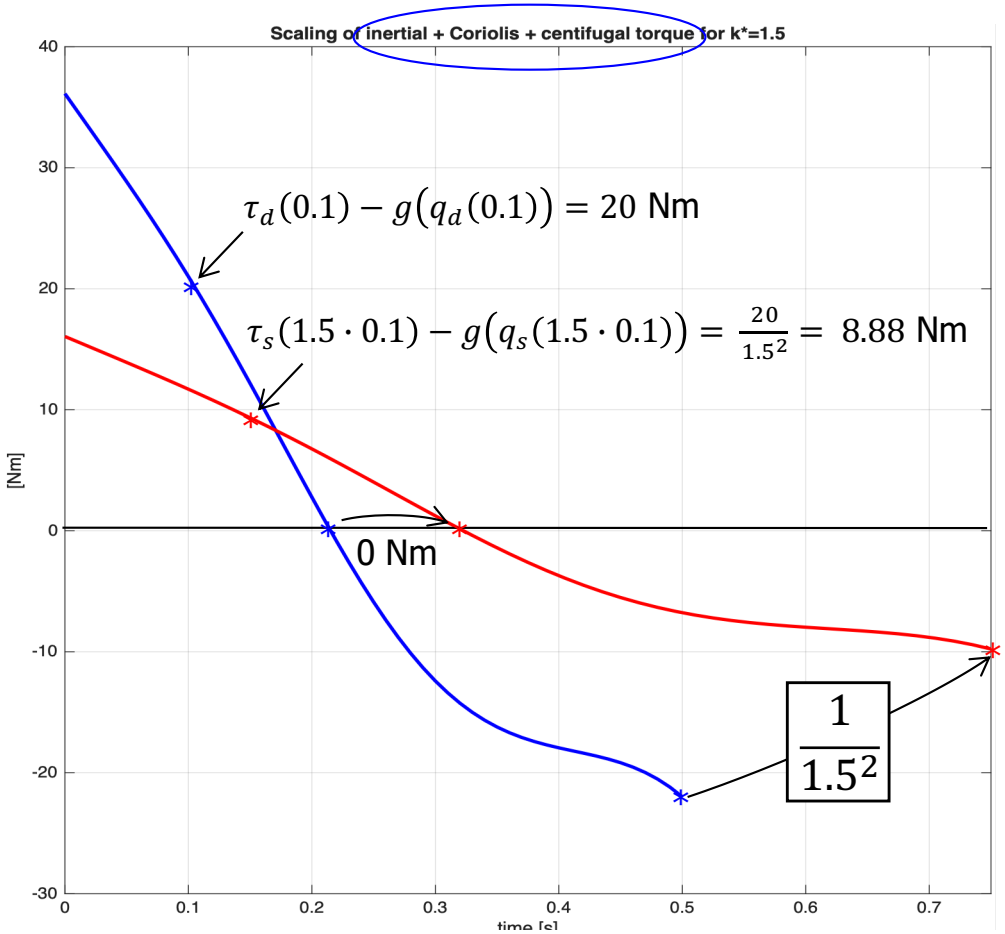
$$k_1 = \frac{36}{16} = 2.25 \quad \Rightarrow \quad k^* = \sqrt{2.25} = 1.5$$

Dynamic scaling of trajectories

numerical example



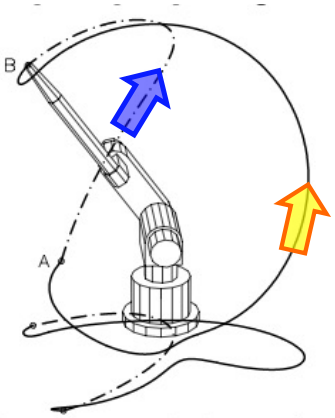
- scaling with $k^* = 1.5$ (slower) $\rightarrow T_s = 0.75$ s



$T = 0.5$ s — $k^* = 1.5$ → $T_s = 0.75$ s

Optimal point-to-point robot motion

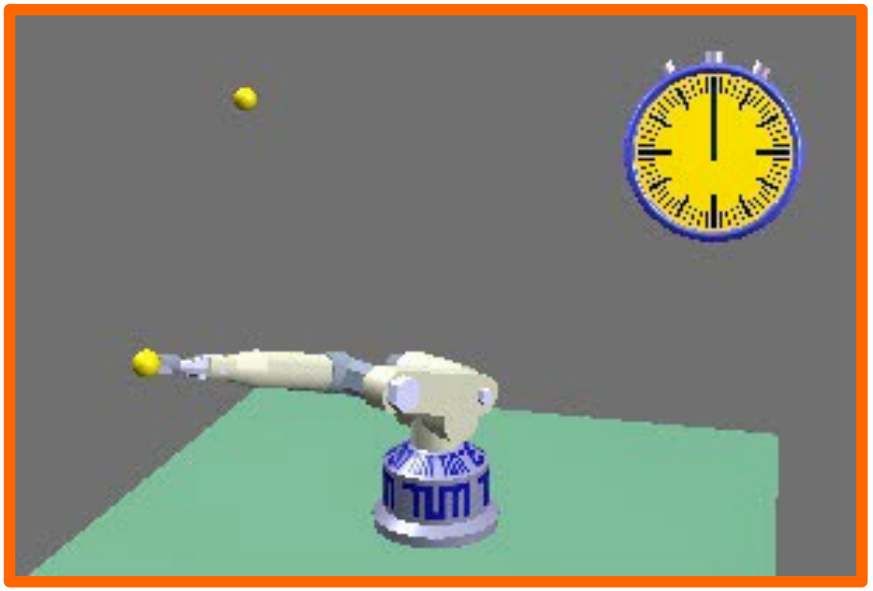
considering the dynamic model



- given the initial (⇒ A) and final (⇒ B) robot configurations (at rest) and the actuator torque bounds, find
  - the minimum-time $T_{\min}$ motion
  - the (global/integral) minimum-energy $E_{\min}$ motion

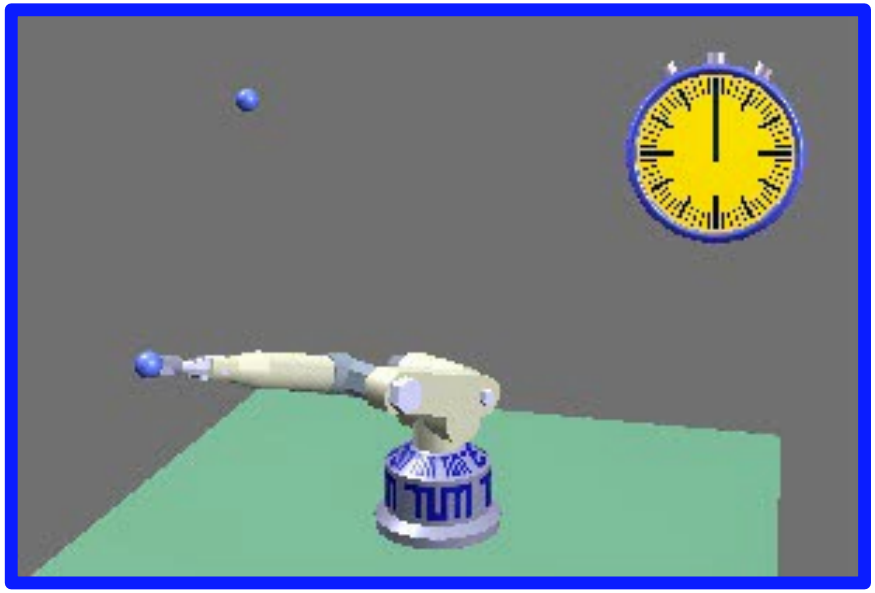
  and the associated command torques needed to execute them
- a complex nonlinear optimization problem solved numerically

video



$T_{\min}$ = 1.32 s, E = 306

video



T = 1.60 s, $E_{\min}$ = 6.14

Optimal point-to-point robot motion

time profiles of different control solutions



minimum-time $T_{\min}$ (with/without state constraints), minimum-energy $E_{\min}$, minimum-power $P_{\min}$

joint velocity bounds

"bang-bang" torques are always saturated (except for singular arcs)

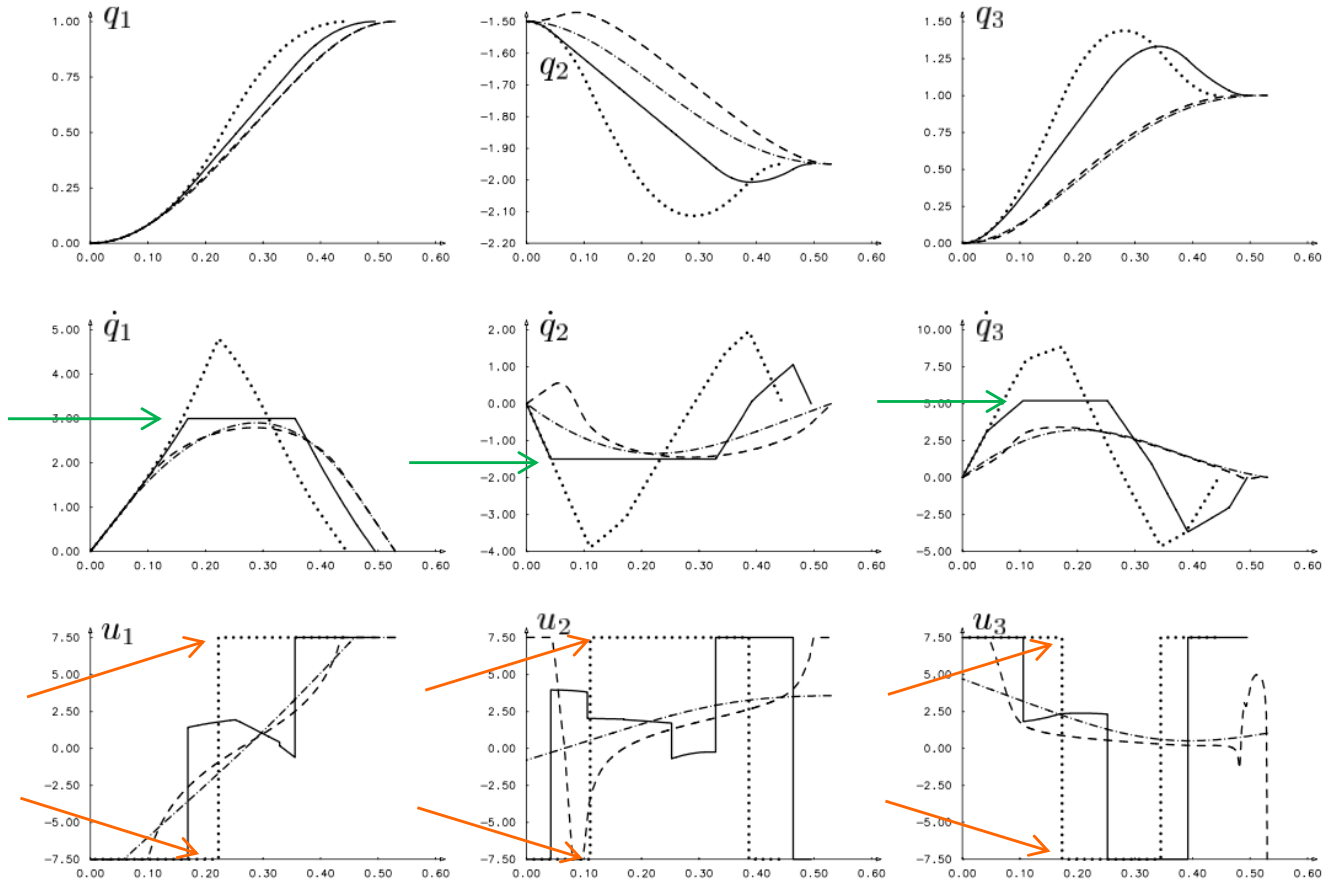


Figure 5: Solution curves of state and control variables for the state constrained time optimal motion (———), the minimum power consumption (– – –), the minimum energy motion (· ·), and the unconstrained time optimal motion (· · · · ·).

O. van Stryk, M. Schlemmer: "Optimal control of the industrial robot Manutec r3" 1994